Probability of Default — Model Card

Version 1.0 · Generated 2026-05-07 · Synthetic-data demonstration · Status: Development

1. Model Overview

Model name	Consumer-loan PD (12-month)
Model type	XGBoost classifier (challenger) + logistic regression (benchmark)
Target variable	default_flag — 90+ days past due in the next 12 months
Use case	Underwriting decisioning and IFRS 9 / CECL provisioning
Population	U.S. consumer term-loan applicants
Output	Probability of default in [0, 1] and decile risk band
Refresh cadence	Quarterly recalibration; full rebuild every 18 months

2. Data

The training population was 33,343 loans (in-time) with an 12.26% default rate. The model was validated on a hold-out validation set (8,336 loans) and a true out-of-time (OOT) sample of 8,321 loans originated in the most recent six months (default rate 12.59%).

Features include borrower demographics, loan characteristics, credit-bureau snapshot at origination (delinquencies, utilization, history length), and one-hot encoded loan purpose and home-ownership flags. Numeric inputs are standard-scaled.

3. Performance (Out-of-Time)

Logistic regression — OOT AUC	0.6390
XGBoost — OOT AUC	0.6260
XGBoost — OOT Gini	0.2520
Score PSI (train vs. OOT)	0.0023 (stable)

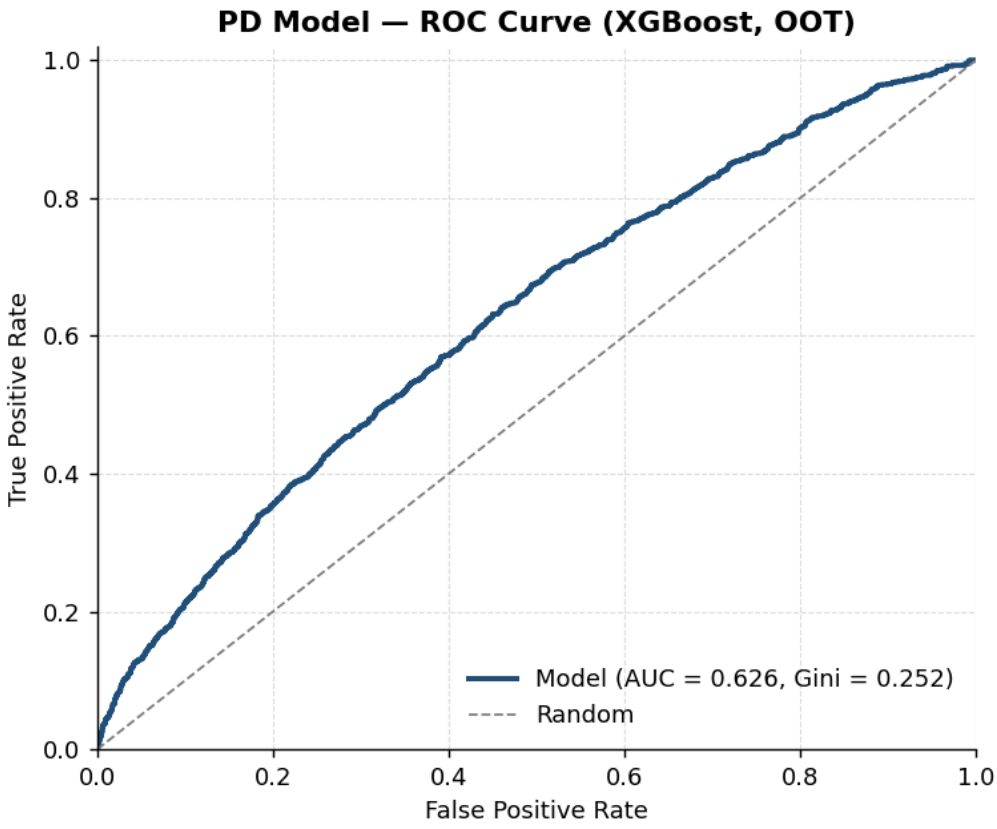

Figure 1. ROC curve on the out-of-time sample. The diagonal is the random-classifier reference.

4. Calibration

Predicted probabilities are systematically high relative to observed default rates. This is the expected consequence of training with **scale_pos_weight** set to balance the classes — the model learns to discriminate but its raw probabilities reflect a 50/50 prior rather than the population base rate. For deployment in an ECL/IFRS-9 context, an isotonic or sigmoid recalibration step is required before the scores are consumed as PDs. Rank-ordering (and therefore the score-band bucketing shown below) is unaffected.

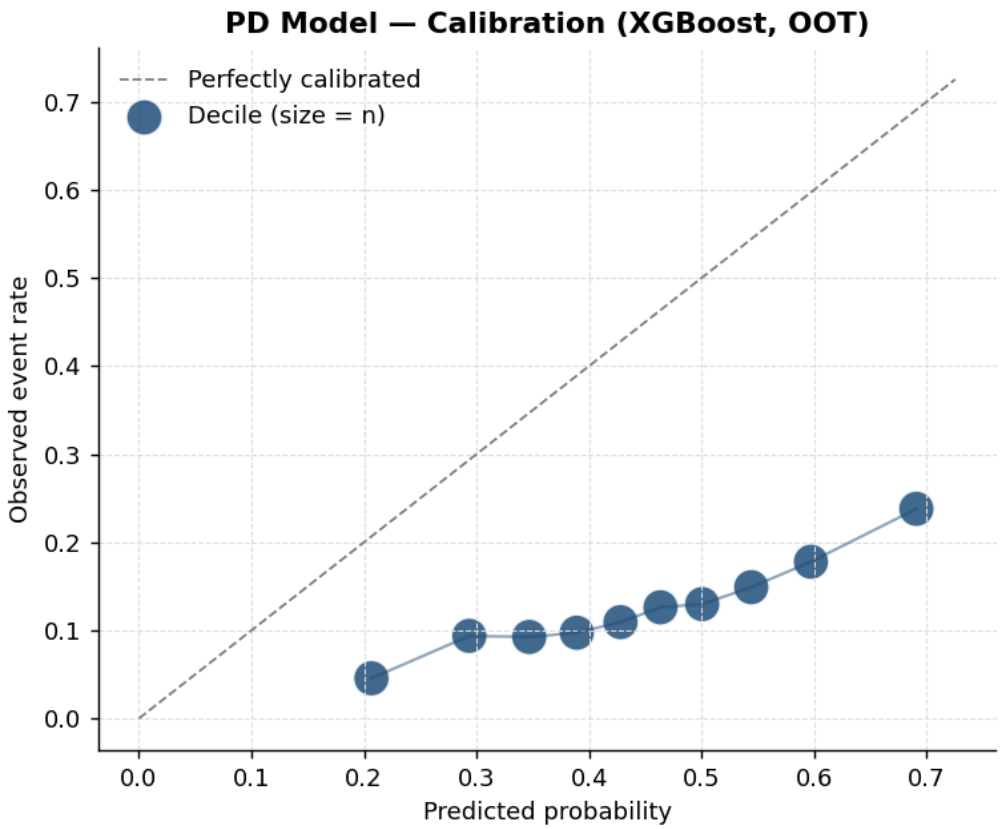

Figure 2. Reliability diagram on OOT. Marker size is proportional to bin count.

5. Decile Lift

Lift relative to the OOT base default rate by predicted-risk decile. The top decile concentrates roughly 2-3× the population default rate; the bottom decile is well below it. Rank-ordering is monotone, which is the property required for cut-off based underwriting decisions.

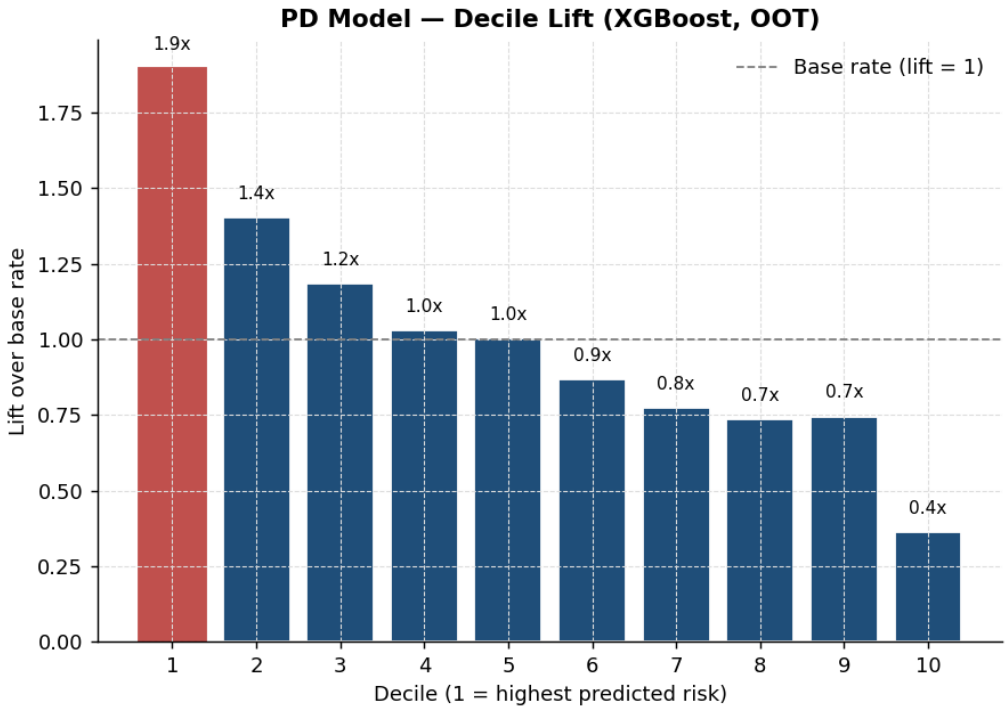

Figure 3. Decile lift on OOT.

6. Feature Importance

Top features by XGBoost gain. Loan-purpose flags, revolving utilization, delinquency history, and credit-history length dominate, consistent with industry experience on consumer term-loan PD models.

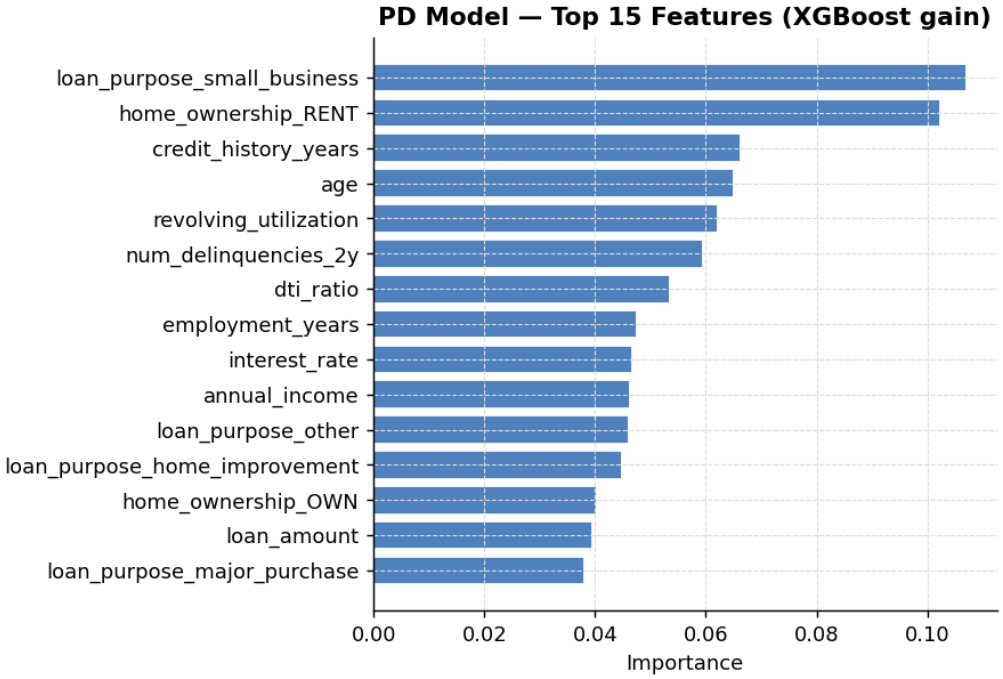

Figure 4. Top 15 features by gain.

7. Limitations

• Trained on synthetic data — coefficients and Gini values are illustrative, not benchmarks.

• Probability calibration is biased high (see Section 4); recalibrate before use as PD.

• No macroeconomic overlay — IFRS 9 forward-looking scenarios would need to be applied externally.

• No segmentation by product type, channel, or vintage; a production version would build separate models or include those as features.

• Bureau attributes are point-in-time at origination; behavioural updates not modeled.

8. Monitoring

Score PSI threshold	< 0.10 stable · 0.10-0.25 watch · > 0.25 investigate
Feature PSI	Monthly; same thresholds, per feature
Performance refresh	Vintage-level Gini/KS as outcomes mature (12-month tag)
Calibration check	Quarterly observed-vs-predicted by decile; recalibrate if MAE > 5 ppt
Override rate	Tracked by underwriter and segment to detect emerging adverse selection

This model card describes a portfolio demonstration trained on synthetic data. It is not an approved production artefact.